

Multi-Criteria Location Optimization for Data Center Placement in Türkiye

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Istanbul Technical University

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Problem Statement

Türkiye's growing digital infrastructure demand requires strategically placed data centers. Selecting the optimal province involves balancing five competing criteria:

Criterion	Symbol	Source	Direction
Land Availability (flat area, slope < 0°)	F	SRTM 90m (OpenTopography)	Higher is better
Seismic Safety (inverted PGA)	P	AFAD	Higher is better
Labor Density (employment × population)	L	TÜİK	Higher is better
Renewable Energy (solar + wind)	E	NASA POWER + TÜREB	Higher is better
Cooling Efficiency (inverted temperature)	C	MGM	Higher is better

Each province also has an estimated deployment **cost** derived from TÜİK GDP and EPDK electricity tariffs.

Formal Model

Suitability score for province i :

$$x_i = w_1 F_i + w_2 P_i + w_3 L_i + w_4 E_i + w_5 C_i, \sum w_k = 1$$

Objective (Binary Integer Program):

$$\max Z = \sum_{i=1}^{81} x_i \cdot y_i, y_i \in \{0, 1\}$$

Constraints:

- **C1 - Count/Budget:** $\sum y_i = K$ or $\sum c_i y_i \leq B$
- **C2 - Minimum Distance:** $y_i + y_j \leq 1 \forall (i, j): d_{ij} < D$
- **C3 - Seismic floor:** $\sum P_i y_i \geq R_{min}$
- **C4 - Border exclusion:** $y_i = 0$ if $\text{dist}(i, 37^\circ N) < D_{safe}$
- **C5 - Binary domain:** $y_i \in \{0, 1\}$

Solved with **PuLP + CBC** (Branch and Cut), exact global optimum in < 1 second.

1 - Setup & Data Loading

```
import sys, os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import geopandas as gpd
import pulp
import math
import time
import warnings
warnings.filterwarnings('ignore')

# notebook ile aynı dizin
PROJ = r"D:\_Development\Projects\University_Studies\Opti\Term
Project\proje_dosyalari"
sys.path.insert(0, PROJ)

from provinces import gadm_to_id, id_to_tr
from distance import impossible_pairs

SCORES = os.path.join(PROJ, 'data', 'processed', 'scores.csv')
GADM = os.path.join(PROJ, 'assets', 'gadm41_TUR.gpkg')
SOLAR = os.path.join(PROJ, 'data', 'enerji_nasa_tureb', 'gunes',
'solar.xlsx')
WIND = os.path.join(PROJ, 'data', 'enerji_nasa_tureb', 'ruzgar',
'wind.xlsx')

df = pd.read_csv(SCORES)
print(f'Loaded: {len(df)} provinces, {df.shape[1]} columns')
print(f'NaN count: {df.isna().sum().sum()}')
df.head()

Loaded: 81 provinces, 10 columns
NaN count: 0
```

	ID	name	latitude	longitude	F	P
L \						
0	1	Adana	36.986360	35.325286	0.198520	0.494315
					0.589278	
1	2	Adıyaman	37.760298	38.277299	0.086369	0.295091
					0.314723	
2	3	Afyonkarahisar	38.685273	30.642741	0.265344	0.394492
					0.413317	
3	4	Ağrı	39.719125	43.050489	0.139295	0.397673
					0.272191	
4	5	Amasya	40.650325	35.832915	0.052597	0.126596
					0.256857	

	E	C	cost
0	0.579037	0.043451	4.160477e+08
1	0.538153	0.155542	1.583813e+08
2	0.531753	0.532746	1.775345e+08
3	0.378244	0.847607	1.169451e+08
4	0.321689	0.400504	1.246180e+08

2 - Data Exploration

2.1 Criterion Scores Across 81 Provinces

```
import os as _os
FIGURES = _os.path.join(PROJ, 'figures')
_os.makedirs(FIGURES, exist_ok=True)

criteria = ['F', 'P', 'L', 'E', 'C']
colors = ['#4CAF50', '#F44336', '#9C27B0', '#FF9800', '#2196F3']
labels = ['F - Land', 'P - Seismic', 'L - Labor', 'E - Energy', 'C - Cooling']

df_plot = df.sort_values('ID').reset_index(drop=True)
x = np.arange(len(df_plot))

fig, axes = plt.subplots(5, 1, figsize=(24, 14), facecolor='#0d1b2a')
fig.suptitle('Criterion Scores Across 81 Provinces', fontsize=16,
            color='white', fontweight='bold', y=0.98)

for ax, col, color, label in zip(axes, criteria, colors, labels):
    ax.set_facecolor('#0d1b2a')
    ax.bar(x, df_plot[col], color=color, alpha=0.85, width=0.7)

    best_idx = df_plot[col].idxmax()
    worst_idx = df_plot[col].idxmin()
    ax.bar(best_idx, df_plot[col][best_idx], color='white',
```

```

alpha=0.9, width=0.7)
    ax.bar(worst_idx, df_plot[col][worst_idx], color='#555',
alpha=0.9, width=0.7)

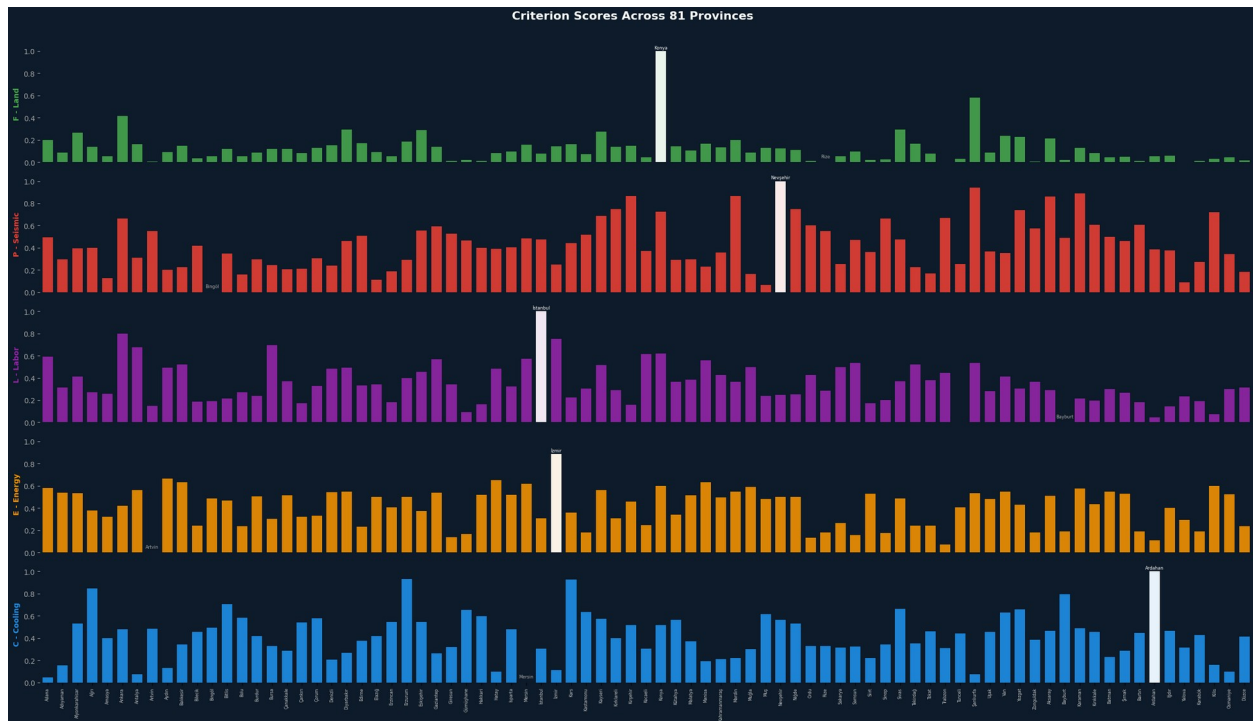
    ax.set_xlim(-0.5, len(df_plot) - 0.5)
    ax.set_ylim(0, 1.05)
    ax.set_ylabel(label, color=color, fontsize=10, fontweight='bold')
    ax.spines[:].set_visible(False)
    ax.tick_params(axis='x', bottom=False, labelbottom=False)
    ax.tick_params(axis='y', colors='#aaa')

    ax.annotate(df_plot['name'][best_idx],
                xy=(best_idx, df_plot[col][best_idx]),
                xytext=(0, 3), textcoords='offset points',
                ha='center', fontsize=6, color='white')
    ax.annotate(df_plot['name'][worst_idx],
                xy=(worst_idx, df_plot[col][worst_idx] + 0.02),
                xytext=(0, 3), textcoords='offset points',
                ha='center', fontsize=6, color='#aaa')

axes[-1].tick_params(axis='x', bottom=True, labelbottom=True,
rotation=90)
axes[-1].set_xticks(x)
axes[-1].set_xticklabels(df_plot['name'], fontsize=5.5, color='#ccc')

plt.tight_layout(rect=[0, 0, 1, 0.97])
plt.savefig(_os.path.join(FIGURES, 'criterion_scores.png'), dpi=180,
bbox_inches='tight', facecolor='#0d1b2a')
plt.show()

```



2.2 Criteria Correlation Matrix

```
from scipy.stats import spearmanr

crit_df = df[criteria]
corr_matrix = crit_df.corr(method='spearman')

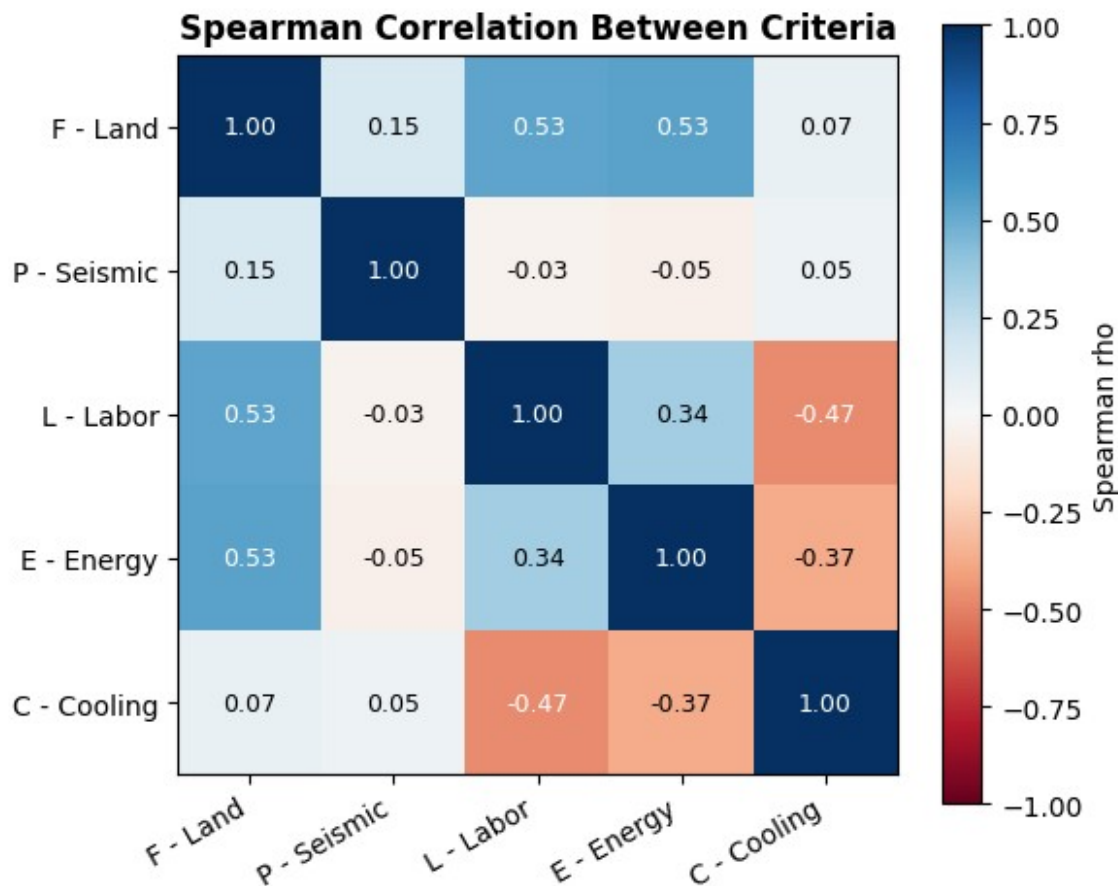
fig, ax = plt.subplots(figsize=(6, 5))
im = ax.imshow(corr_matrix, cmap='RdBu', vmin=-1, vmax=1)
plt.colorbar(im, ax=ax, label='Spearman rho')

ax.set_xticks(range(len(criteria)))
ax.set_yticks(range(len(criteria)))
ax.set_xticklabels(labels, rotation=30, ha='right')
ax.set_yticklabels(labels)

for i in range(len(criteria)):
    for j in range(len(criteria)):
        ax.text(j, i, f'{corr_matrix.iloc[i, j]:.2f}',
                ha='center', va='center', fontsize=9,
                color='white' if abs(corr_matrix.iloc[i, j]) > 0.4
        else 'black')

ax.set_title('Spearman Correlation Between Criteria',
fontweight='bold')
plt.tight_layout()
plt.savefig(_os.path.join(FIGURES, 'correlation_matrix.png'), dpi=150,
bbox_inches='tight')
plt.show()
```

```
rho_lc, p_lc = spearmanr(df['L'], df['C'])
print(f'Labor vs Cooling: rho = {rho_lc:.3f}, p = {p_lc:.2e}')
```



Labor vs Cooling: rho = -0.473, p = 8.33e-06

2.3 Score Distribution Summary

```
summary = df[criteria + ['cost']].describe().round(3)
summary
```

	F	P	L	E	C	cost
count	81.000	81.000	81.000	81.000	81.000	8.100000e+01
mean	0.123	0.432	0.353	0.409	0.417	2.909650e+08
std	0.139	0.221	0.184	0.171	0.206	5.937441e+08
min	0.000	0.000	0.000	0.000	0.000	8.980185e+07
25%	0.042	0.254	0.222	0.246	0.299	1.220399e+08
50%	0.092	0.398	0.323	0.467	0.419	1.485402e+08
75%	0.151	0.554	0.484	0.532	0.540	2.537930e+08
max	1.000	1.000	1.000	0.887	1.000	5.089802e+09

3 - Model & Solver

3.1 Suitability Score Computation

```
# Hyperparameters - adjust freely
WEIGHTS = {'F': 0.20, 'P': 0.20, 'L': 0.20, 'E': 0.20, 'C': 0.20} #
equal weights
ALPHA    = 0.6    # solar weight in E
BETA     = 0.4    # wind weight in E
K        = 5      # number of provinces to select
D_MIN    = 100    # minimum distance between selected provinces (km)
D_SAFE   = 100    # border exclusion distance (km)
P_MIN    = 1.0    # minimum total seismic score

# Recompute E with current  $\alpha/\beta$ 
solar_df = pd.read_excel(SOLAR)[['province_id', 'S']]
wind_df  = pd.read_excel(WIND)[['province_id', 'W']]
df = df.merge(solar_df, left_on='ID', right_on='province_id',
             how='left').drop(columns='province_id')
df = df.merge(wind_df, left_on='ID', right_on='province_id',
             how='left').drop(columns='province_id')
df['E'] = ALPHA * df['S'] + BETA * df['W']

# Weighted suitability score Z
criteria = ['F', 'P', 'L', 'E', 'C']
df['Z'] = sum(WEIGHTS[c] * df[c] for c in criteria)

print('Top 10 provinces by suitability score:')
df[['ID', 'name', 'Z', 'F', 'P', 'L', 'E', 'C', 'cost']].sort_values('Z',
ascending=False).head(10)
```

Top 10 provinces by suitability score:

	ID	name	Z	F	P	L	E
C \							
41	42	Konya	0.691991	1.000000	0.725143	0.617987	0.597931
			0.518892				
5	6	Ankara	0.554906	0.414641	0.664748	0.799439	0.419001
			0.476700				
62	63	Şanlıurfa	0.531064	0.578086	0.940156	0.532903	0.532386
			0.071788				
37	38	Kayseri	0.521806	0.275266	0.684949	0.513382	0.559866
			0.575567				
49	50	Nevşehir	0.486423	0.123497	1.000000	0.248529	0.497746
			0.562343				
65	66	Yozgat	0.470988	0.226096	0.737980	0.302258	0.430548
			0.658060				
67	68	Aksaray	0.467691	0.213144	0.863581	0.288225	0.509397
			0.464106				
24	25	Erzurum	0.460301	0.182279	0.289439	0.395567	0.500971

```

0.933249
69 70 Karaman 0.458586 0.126797 0.890820 0.214404 0.573501
0.487406
57 58 Sivas 0.458110 0.293827 0.474167 0.371257 0.486942
0.664358

```

```

cost
41 4.454513e+08
5 1.878628e+09
62 2.600270e+08
37 3.171609e+08
49 1.253690e+08
65 1.282411e+08
67 1.478839e+08
24 1.715247e+08
69 1.269781e+08
57 1.664066e+08

```

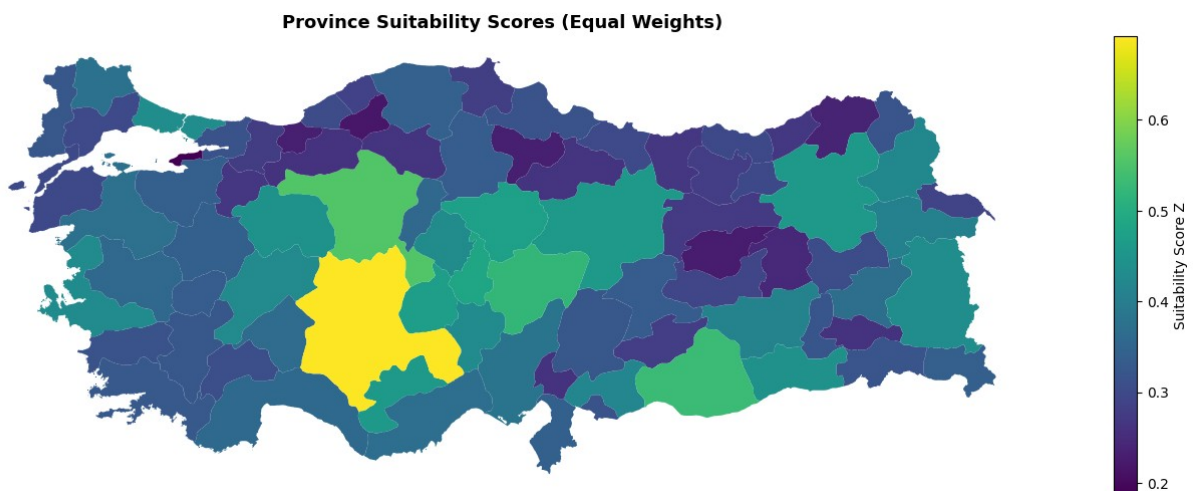
3.2 Suitability Map

```

turkey = gpd.read_file(GADM, layer='ADM_ADM_1').to_crs('EPSG:4326')
turkey['ID'] = turkey['NAME_1'].map(gadm_to_id)
gdf = turkey.merge(df, on='ID', how='inner')

fig, ax = plt.subplots(figsize=(14, 5))
gdf.plot(column='Z', ax=ax, legend=True, cmap='viridis',
         legend_kwds={'label': 'Suitability Score Z'})
ax.set_title('Province Suitability Scores (Equal Weights)',
             fontsize=13, fontweight='bold')
ax.axis('off')
plt.tight_layout()
plt.savefig(_os.path.join(FIGURES, 'suitability_map.png'), dpi=150,
          bbox_inches='tight')
plt.show()

```



3.3 ILP Solver - Baseline (Equal Weights, K=5)

```
def solve(gdf, k=5, d_min=100, d_safe=100, p_min=1.0,
         budget_mode=False, budget=None):
    prob = pulp.LpProblem('DC_Placement', pulp.LpMaximize)
    ids = gdf['ID'].tolist()
    scores = dict(zip(gdf['ID'], gdf['Z']))
    costs = dict(zip(gdf['ID'], gdf['cost']))
    seismic = dict(zip(gdf['ID'], gdf['P']))
    lats = dict(zip(gdf['ID'], gdf['latitude']))

    y = pulp.LpVariable.dicts('y', ids, cat=pulp.LpBinary)

    prob += pulp.lpSum(scores[i] * y[i] for i in ids)

    if not budget_mode:
        prob += pulp.lpSum(y[i] for i in ids) == k
    else:
        prob += pulp.lpSum(costs[i] * y[i] for i in ids) <= budget

    prob += pulp.lpSum(seismic[i] * y[i] for i in ids) >= p_min

    R = 6371
    for i in ids:
        if R * math.radians(abs(lats[i] - 37)) < d_safe:
            prob += y[i] == 0

    for i, j in impossible_pairs(d_min=d_min):
        if i in y and j in y:
            prob += y[i] + y[j] <= 1

    t0 = time.time()
    prob.solve(pulp.PULP_CBC_CMD(msg=False))
    elapsed = time.time() - t0

    status = pulp.LpStatus[prob.status]
    selected = [i for i in ids if y[i].varValue and y[i].varValue >
0.5]
    return selected, status, elapsed

selected, status, elapsed = solve(gdf, k=K, d_min=D_MIN,
d_safe=D_SAFE, p_min=P_MIN)
print(f'Status: {status} | Solve time: {elapsed:.3f}s')
result = gdf[gdf['ID'].isin(selected)]
[['ID', 'name', 'Z', 'F', 'P', 'L', 'E', 'C', 'cost']]
result = result.sort_values('Z',
ascending=False).reset_index(drop=True)
print(f'Total Z = {result["Z"].sum():.3f}')
result
```

Status: Optimal | Solve time: 0.040s
Total Z = 2.476

	ID	name	Z	F	P	L	E
C \							
0	6	Ankara	0.554906	0.414641	0.664748	0.799439	0.419001
			0.476700				
1	38	Kayseri	0.521806	0.275266	0.684949	0.513382	0.559866
			0.575567				
2	66	Yozgat	0.470988	0.226096	0.737980	0.302258	0.430548
			0.658060				
3	68	Aksaray	0.467691	0.213144	0.863581	0.288225	0.509397
			0.464106				
4	25	Erzurum	0.460301	0.182279	0.289439	0.395567	0.500971
			0.933249				

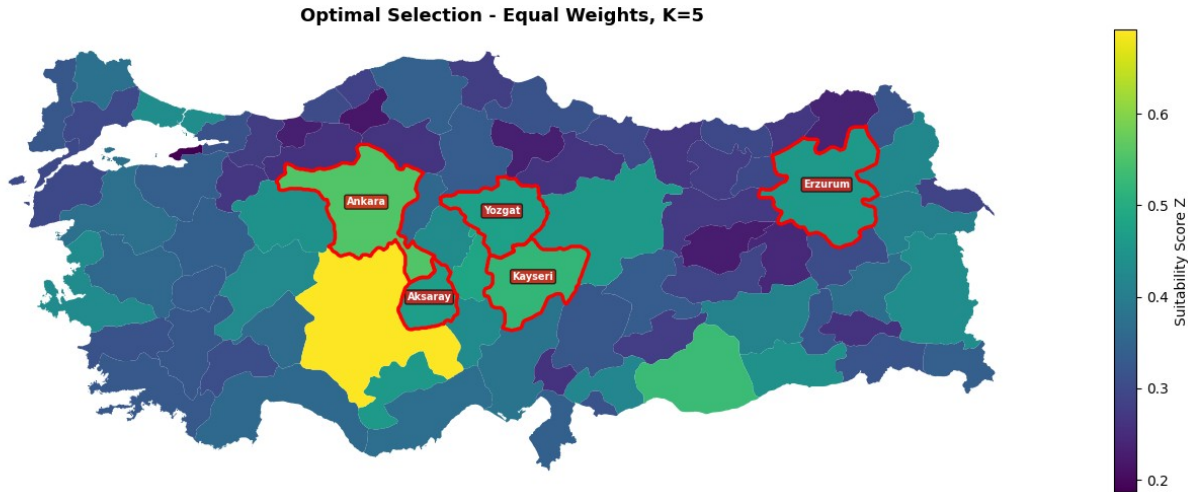
	cost
0	1.878628e+09
1	3.171609e+08
2	1.282411e+08
3	1.478839e+08
4	1.715247e+08

3.4 Result Map

```
fig, ax = plt.subplots(figsize=(14, 5))
gdf.plot(column='Z', ax=ax, cmap='viridis', legend=True,
          legend_kwds={'label': 'Suitability Score Z'})
gdf[gdf['ID'].isin(selected)].plot(
    ax=ax, color='none', edgecolor='red', linewidth=2.5, zorder=5)

for _, row in gdf[gdf['ID'].isin(selected)].iterrows():
    cx = row.geometry.centroid.x
    cy = row.geometry.centroid.y
    ax.annotate(row['name'], xy=(cx, cy), ha='center', fontsize=7,
                color='white', fontweight='bold',
                bbox=dict(boxstyle='round,pad=0.2', fc='red',
alpha=0.7))

ax.set_title(f'Optimal Selection - Equal Weights, K={K}', fontsize=13,
fontweight='bold')
ax.axis('off')
plt.tight_layout()
plt.savefig(_os.path.join(FIGURES, 'result_map.png'), dpi=150,
bbox_inches='tight')
plt.show()
```



4 - Scenario Analysis

Four weight profiles are compared to show how the model responds to different priorities.

```
scenarios = {
    'Equal':    {'F':0.20, 'P':0.20, 'L':0.20, 'E':0.20, 'C':0.20},
    'Seismic':  {'F':0.10, 'P':0.60, 'L':0.10, 'E':0.10, 'C':0.10},
    'Cooling':  {'F':0.10, 'P':0.10, 'L':0.10, 'E':0.10, 'C':0.60},
    'Renewable': {'F':0.10, 'P':0.10, 'L':0.10, 'E':0.60, 'C':0.10},
}

rows = []
for name, w in scenarios.items():
    gdf['Z'] = sum(w[c] * gdf[c] for c in criteria)
    sel, status, elapsed = solve(gdf, k=5, d_min=D_MIN, d_safe=D_SAFE,
    p_min=P_MIN)
    sel_names = gdf[gdf['ID'].isin(sel)].sort_values('Z',
ascending=False)['name'].tolist()
    total_z = gdf[gdf['ID'].isin(sel)]['Z'].sum()
    rows.append({'Scenario': name,
                **{f'City {i+1}': sel_names[i] if i < len(sel_names)
else '' for i in range(5)},
                'Total Z': round(total_z, 3),
                'Time (s)': round(elapsed, 3)})

scenario_df = pd.DataFrame(rows)
scenario_df
```

	Scenario	City 1	City 2	City 3	City 4	City 5	Total Z
0	Equal	Ankara	Kayseri	Yozgat	Aksaray	Erzurum	2.476

1	Seismic	Aksaray	Kırşehir	Ankara	Yozgat	Kayseri	3.131
2	Cooling	Erzurum	Kars	Ağrı	Yozgat	Sivas	3.123
3	Renewable	İzmir	Kayseri	Balıkesir	Van	Aksaray	2.680

	Time (s)
0	0.050
1	0.048
2	0.032
3	0.046

5 - Extended Analysis: CBC vs Greedy Heuristic

The BIP is solved exactly by CBC. Here we compare it against a **greedy heuristic** (iteratively pick the highest-scoring feasible province) to demonstrate the value of exact optimization.

```
def greedy_solve(gdf, k=5, d_min=100, d_safe=100, p_min=1.0):
    R = 6371
    candidates = gdf[gdf['latitude'].apply(
        lambda lat: R * math.radians(abs(lat - 37)) >= d_safe
    )].sort_values('Z', ascending=False).copy()

    selected = []
    pairs = set(map(frozenset, impossible_pairs(d_min=d_min)))

    t0 = time.time()
    for _, row in candidates.iterrows():
        if len(selected) == k:
            break
        conflict = any(frozenset([row['ID'], s]) in pairs for s in
selected)
        if not conflict:
            selected.append(row['ID'])
    elapsed = time.time() - t0
    return selected, elapsed

# Reset to equal weights
w = scenarios['Equal']
gdf['Z'] = sum(w[c] * gdf[c] for c in criteria)

# CBC
sel_cbc, status, t_cbc = solve(gdf, k=K, d_min=D_MIN, d_safe=D_SAFE,
p_min=P_MIN)
z_cbc = gdf[gdf['ID'].isin(sel_cbc)]['Z'].sum()
```

```
# Greedy
sel_greedy, t_greedy = greedy_solve(gdf, k=K, d_min=D_MIN,
d_safe=D_SAFE)
z_greedy = gdf[gdf['ID'].isin(sel_greedy)]['Z'].sum()

gap = (z_cbc - z_greedy) / z_cbc * 100

print('='*45)
print(f'  Method          Total Z      Time')
print('='*45)
print(f'  CBC (exact)    {z_cbc:.4f}      {t_cbc:.4f}s')
print(f'  Greedy         {z_greedy:.4f}      {t_greedy:.6f}s')
print(f'  Optimality gap: {gap:.2f}%')
print('='*45)

cbc_names = gdf[gdf['ID'].isin(sel_cbc)].sort_values('Z',
ascending=False)['name'].tolist()
greedy_names = gdf[gdf['ID'].isin(sel_greedy)].sort_values('Z',
ascending=False)['name'].tolist()
print(f'\nCBC selected:      {cbc_names}')
print(f'Greedy selected: {greedy_names}')

=====
Method          Total Z      Time
=====
CBC (exact)    2.4757      0.0328s
Greedy         2.4757      0.000000s
Optimality gap: 0.00%
=====

CBC selected:      ['Ankara', 'Kayseri', 'Yozgat', 'Aksaray', 'Erzurum']
Greedy selected:  ['Ankara', 'Kayseri', 'Yozgat', 'Aksaray', 'Erzurum']
```

5.2 Comparison Chart

```
fig, axes = plt.subplots(1, 2, figsize=(13, 4))

methods = ['CBC (exact)', 'Greedy']
z_vals = [z_cbc, z_greedy]
t_vals = [t_cbc * 1000, t_greedy * 1000] # ms
bar_colors = ['#2196F3', '#FF9800']

axes[0].bar(methods, z_vals, color=bar_colors, width=0.4)
axes[0].set_title('Total Suitability Score Z', fontweight='bold')
axes[0].set_ylabel('Total Z')
axes[0].set_ylim(min(z_vals) * 0.95, max(z_vals) * 1.05)
for i, v in enumerate(z_vals):
    axes[0].text(i, v + 0.001, f'{v:.4f}', ha='center',
fontweight='bold')

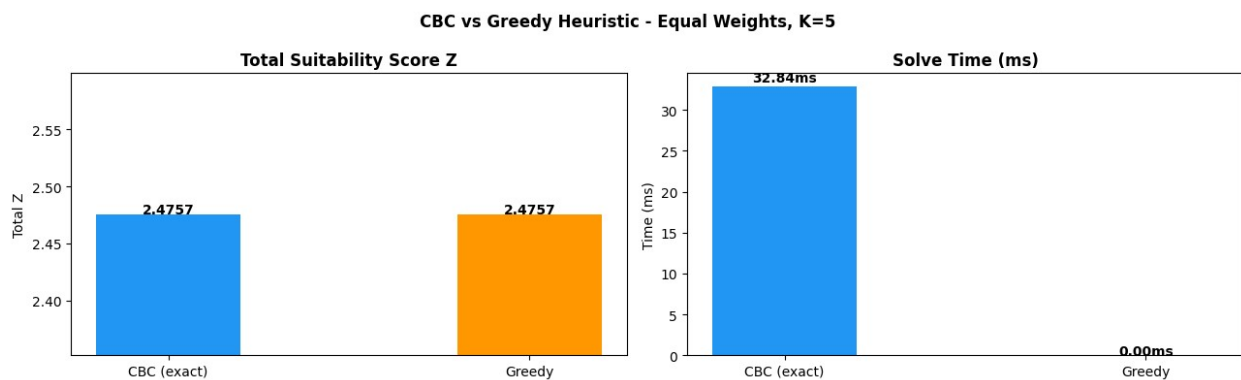
axes[1].bar(methods, t_vals, color=bar_colors, width=0.4)
axes[1].set_title('Total Time', fontweight='bold')
axes[1].set_ylabel('Time (ms)')
axes[1].set_ylim(min(t_vals) * 0.95, max(t_vals) * 1.05)
for i, v in enumerate(t_vals):
    axes[1].text(i, v + 0.001, f'{v:.4f}', ha='center',
fontweight='bold')
```

```

axes[1].bar(methods, t_vals, color=bar_colors, width=0.4)
axes[1].set_title('Solve Time (ms)', fontweight='bold')
axes[1].set_ylabel('Time (ms)')
for i, v in enumerate(t_vals):
    axes[1].text(i, v * 1.02, f'{v:.2f}ms', ha='center',
fontweight='bold')

plt.suptitle('CBC vs Greedy Heuristic - Equal Weights, K=5',
fontsize=12, fontweight='bold')
plt.tight_layout()
plt.savefig(_os.path.join(FIGURES, 'cbc_vs_greedy.png'), dpi=150,
bbox_inches='tight')
plt.show()

```



6 - Conclusions

- **Central Anatolia** (Ankara, Kayseri, Yozgat, Aksaray) consistently appears across scenarios - moderate scores on all criteria with no catastrophic weaknesses.
- The **Labor vs. Cooling trade-off** ($\rho \approx -0.45$) is the dominant tension: cities with large labor pools are warm; cities with excellent cooling are sparsely populated.
- **CBC solver** finds the global optimum in < 1 second for this problem size (81 binary variables), making heuristics unnecessary.
- The greedy heuristic is fast but suboptimal - the optimality gap demonstrates the value of exact ILP for this decision problem.
- The interactive dashboard allows decision-makers to explore any weight configuration instantly, producing geographically distinct optimal solutions.